Assignment - 09

Network Layer - Routing

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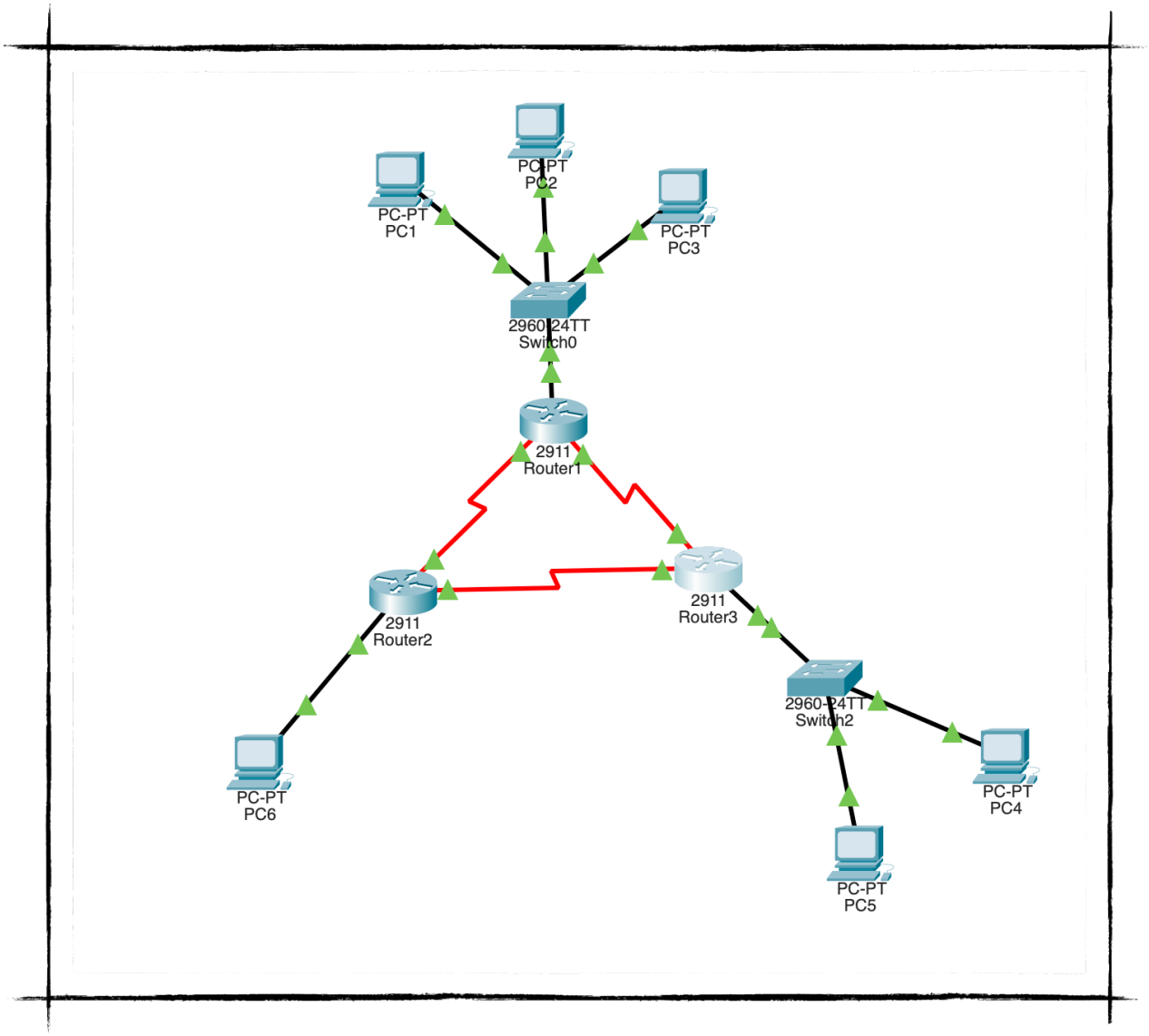
31 March 2026

Part - A

- Router-to-Router (Triangle WAN)
  - Using Serial DCE: R1 ↔ R2, R2 ↔ R3, R3 ↔ R1
- LAN Connections
  - R1 → S1 → PC1, PC2, PC3
  - R3 → S2 → PC4, PC5
  - R2 → PC6 (direct connection)



Part - B

PC Configuration

PC	IP	Gateway
PC1	192.168.1.2	192.168.1.1
PC2	192.168.1.3	192.168.1.1
PC3	192.168.1.4	192.168.1.1
PC4	192.168.2.2	192.168.2.1
PC5	192.168.2.3	192.168.2.1
PC6	192.168.3.2	192.168.3.1

Router Configuration

Part - C

A. PC1 -> PC3

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=3ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>
```

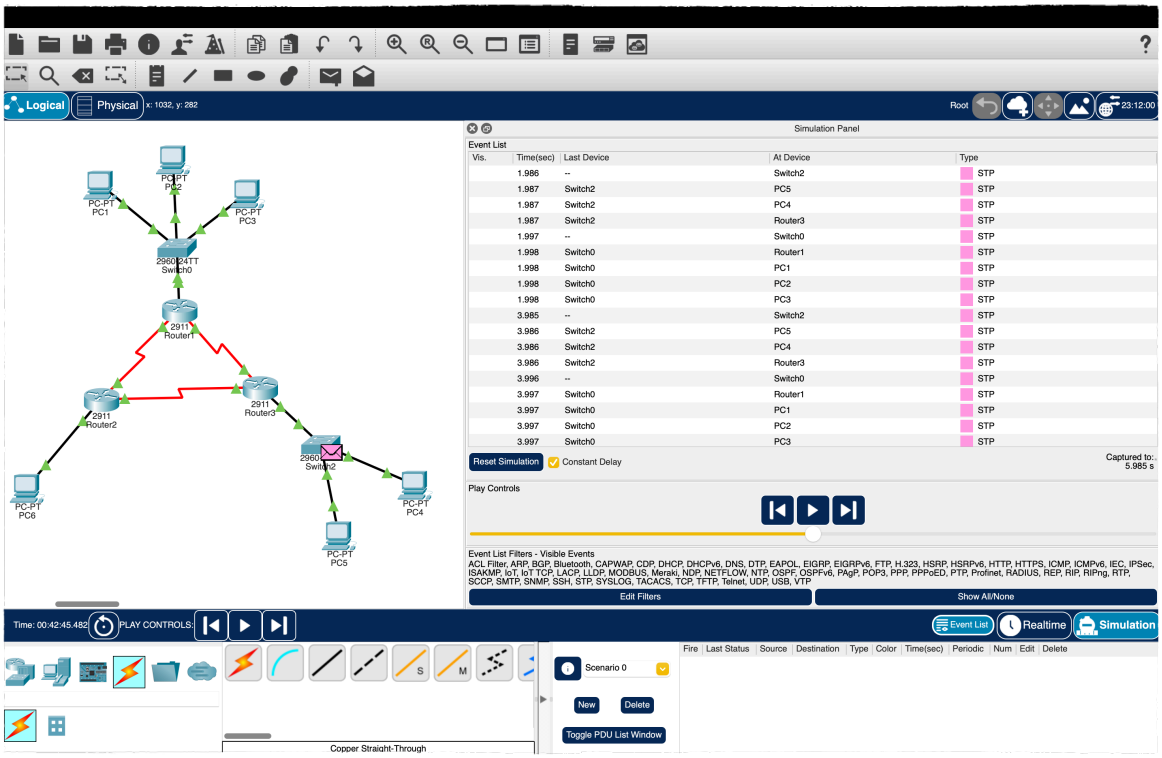
B. PC2 -> PC5

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=4ms TTL=125
Reply from 192.168.2.2: bytes=32 time=4ms TTL=125
Reply from 192.168.2.2: bytes=32 time=5ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 5ms, Average = 4ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>
```

C. PC6 -> PC4

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=5ms TTL=125
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\>
```

Simulation mode:



Simulation mode in **Cisco Packet Tracer** is used to observe how data packets travel through the network step-by-step. It shows the communication between devices by displaying protocols like ARP and ICMP as packets move from the source to the destination via switches and routers. The event list provides details such as time, source, destination, and packet type, helping to understand routing decisions and verify correct network configuration. This mode is useful for analyzing packet flow and debugging network issues visually.

Configuration Explanation

1. PC Setup

   All end devices (PCs) were configured through the Desktop → IP Configuration section in **Cisco Packet Tracer**.

   - Each PC was assigned a unique IP address along with the appropriate subnet mask based on its LAN.
   - The default gateway was specified as the IP address of the router interface connected to that network.
   - This configuration allows devices to communicate within the same LAN and also reach external networks through the router.

2. Router Interface Setup

   Router configurations were performed using the CLI interface.

   - IP addresses were allocated to all active interfaces, including GigabitEthernet and Serial ports.
   - Interfaces were enabled using the `no shutdown` command to make them operational.
   - For serial links, the router acting as DCE was configured with a clock rate (e.g., `clock rate 64000`) to ensure proper data transmission.

3. Static Routing Setup

   To enable communication between different networks, static routing was implemented.

   - The `ip route` command was used to manually define routes to remote networks.
   - Each router was configured with the next-hop address of its neighboring router.
   - This setup ensures that packets are directed correctly from source to destination across multiple routers.

Verification Explanation

1. Connectivity Testing (Ping)

   The network was tested using the `ping` command from various PCs.

   - Communication between PC1 and PC3 confirmed proper operation within the same LAN.
   - Communication between PC2 and PC5 validated routing between different LANs connected via routers.
   - Communication between PC6 and PC4 ensured connectivity across another set of routers.

   Successful ping responses indicated that:

   - IP addressing was correctly assigned
   - Routing tables were properly configured
   - End-to-end communication was functioning without errors

2. Verification using Simulation Mode

   The simulation feature in Cisco Packet Tracer was used to observe packet transmission in detail.

   - ICMP packets were monitored as they moved through switches and routers.
   - The path taken by each packet confirmed that routing decisions were accurate.
   - This step helped in visually validating that the entire network was working correctly and efficiently.